

TS2 - QUESTION BANK (With Solution)

ACN (315321)

2marks

1. List out services of UDP.
2. List out port numbers for HTTP, DNS, SMTP and IMAP.
3. Define World Wide Web.
4. Define Anonymous FTP.
5. Define Static and Dynamic Web documents.
6. What is the Application are for RTP (Real Time Transfer Protocol).
7. What is the main idea behind Software Defined Networking (SDN)?
8. List out features of TCP.
9. What is Edge computing? List out any two challenges.

4marks

1. Draw and explain TCP segment structure.
2. Explain the TCP Three-Way Handshake process used to establish a connection.
3. Explain working of Transport Layer Security (TLS) protocol.
4. Explain Error Control in TCP.
5. Differentiate between POP3 and IMAP.
6. Explain working of TELNET with help of diagram.
7. Explain working of PGP with help of diagram.
8. Explain SDN architecture.
9. Explain Voice-over Internet Protocol (VoIP), How VoIP works?

6marks

1. Explain in detail TCP Flow Control with the help of Receiver Window and Sender Window diagrams.
2. Define **cwnd**. Explain various congestion control methods of transport layer.
3. Describe FTP. Explain communication over control connection and data connection in FTP.
4. List out SMTP commands. Use an example to show how a mail is sent using SMTP commands.
5. Illustrate the importance of Network Function Virtualization (NFV) with example.

TS2 - QUESTION BANK SOLUTION

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2marks

1. List out services of UDP.

- Process-to-Process Communication
- Message-Oriented Protocol
- Connectionless Services
- Low Overhead
- Fast Service
- No Acknowledgment or Handshaking
- No Flow Control and No Error Control

2. List out port numbers for HTTP, DNS, SMTP and IMAP.

Protocol	Port No
HTTP	80
DNS	53
SMTP	25
IMAP	143

3. Define World Wide Web.

WWW is a repository of information linked together from points all over the world. The WWW has a unique combination of flexibility, portability, and user-friendly features that distinguish it from other services provided by the Internet.

4. Define Anonymous FTP.

Anonymous FTP (File Transfer Protocol) is a method that allows users to access and download files from an FTP server without needing a personal username or password.

Instead of requiring an account, users log in using the username “anonymous” and usually enter their email address (or nothing) as the password.

5. Define Static and Dynamic Web documents.

Static Web Documents

Static web documents are fixed content pages that are pre-stored on the web server and sent to the client exactly as they are.

They do not change unless the developer manually updates them.

Dynamic Web Documents

Dynamic web documents are generated by the web server at runtime based on user requests or data.

They can change depending on inputs, database content, or session details.

6. What is the Application are for RTP (Real Time Transfer Protocol).

- Video Conferencing
- Voice over IP (VoIP)
- Live Streaming
- Online Gaming
- Internet Radio

7. What is the main idea behind Software Defined Networking (SDN)?

The main idea of SDN is to separate the control plane from the data plane in networking devices.

This means that the control logic (controller) is centralized, while switches and routers only forward packets based on the controller's instructions.

8. List out features of TCP.

- Numbering System
- Sequence Number
- Flow Control
- Error Control
- Congestion Control

9. What is Edge computing? List out any two challenges.

Edge computing is a computing approach in which data processing and computation are performed close to the location where data is generated, rather than sending all data to a distant centralized cloud or data center.

1. Security and Privacy

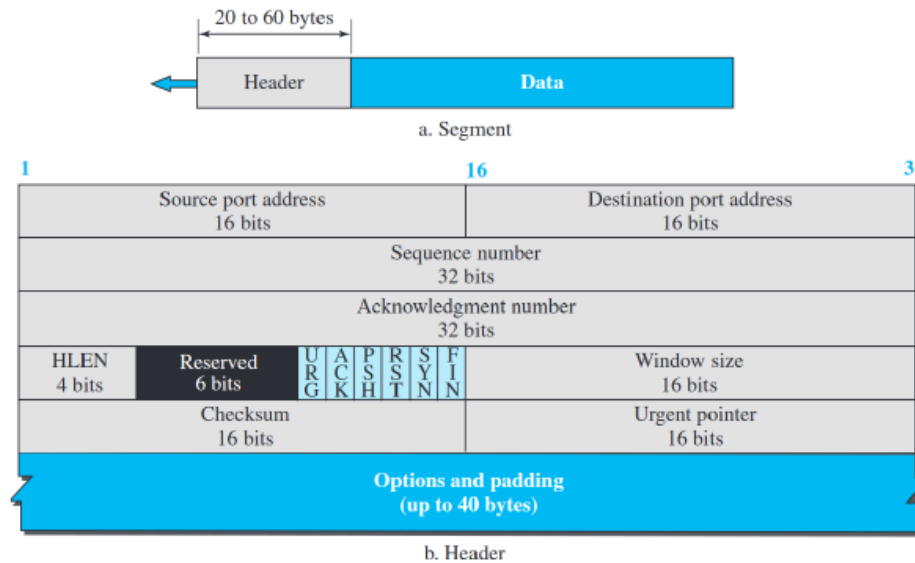
Edge devices are distributed in many locations and may be physically accessible, making them more difficult to protect.

2. Resource Limitations

Edge devices generally have limited CPU, memory, storage, and power compared with centralized cloud servers.

4marks

1. TCP Segment Format



Source Port Number

This is a 16-bit field that defines the port number of the application program in the host that is sending the segment.

Destination Port Number

This is a 16-bit field that defines the port number of the application program in the host that is receiving the segment.

Sequence Number

This 32-bit field defines the number assigned to the first byte of data contained in this segment.

Acknowledgment Number

This 32-bit field defines the byte number that the receiver of the segment is expecting to receive from the sender.

Header Length

This 4-bit field indicates the number of 4-byte words in the TCP header. The length of the header can be between 20 and 60 bytes.

Therefore, the value of this field can be between 5 ($5 \times 4 = 20$) and 15 ($15 \times 4 = 60$).

if $HLEN = 0101 = 5$ in decimal

$5 \times 4B = 20B$ (Minimum length of TCP Header)

if $HLEN = 1111 = 15$ in decimal

$15 \times 4B = 60B$ (Maximum length of TCP Header)

Reserved

This is a 6-bit field reserved for future use.

Window Size

This 16 bit field defines the size of the window, in bytes, that the receiver must maintain.

maximum size of the window is 65,535 bytes (2^{16}).

This value is normally referred to as the receiving window (rwnd) and is determined by the receiver.

The sender can send number of bytes equal to window size, to receiver without worrying acknowledgement because receiver have that size (window size) buffer to save the data bytes.

Checksum

This 16-bit field contains the checksum. The calculation of the checksum for TCP follows the same procedure as the one described for UDP (refer slide 6 and 7).

The inclusion of the checksum for TCP is mandatory.

The same pseudoheader (as UDP), serving the same purpose, is added to the segment.

For the TCP pseudoheader, the value for the protocol field is 6.

Urgent Pointer (16bit)

This 16-bit field, which is valid only if the urgent flag is set. It is used when the segment contains urgent data. It defines the number that must be added to the sequence number to obtain the number of the last urgent byte.

Options

There can be up to 40 bytes of optional information in the TCP header.

2. Explain the TCP Three-Way Handshake process used to establish a connection.

TCP Connection Establishment (Three Way Handshaking)

Three-Way Handshaking The connection establishment in TCP is called three-way handshaking.

The three steps in this phase are as follows.

SYN

The client sends the first segment, a SYN segment, in which only the SYN flag is set. This segment is for synchronization of sequence numbers.

SYN + ACK

The server sends the second segment, a SYN + ACK segment, with 2 flag bits set.

ACK

The client sends the third segment. This is just an ACK segment. It guarantees the completion of three way handshaking.

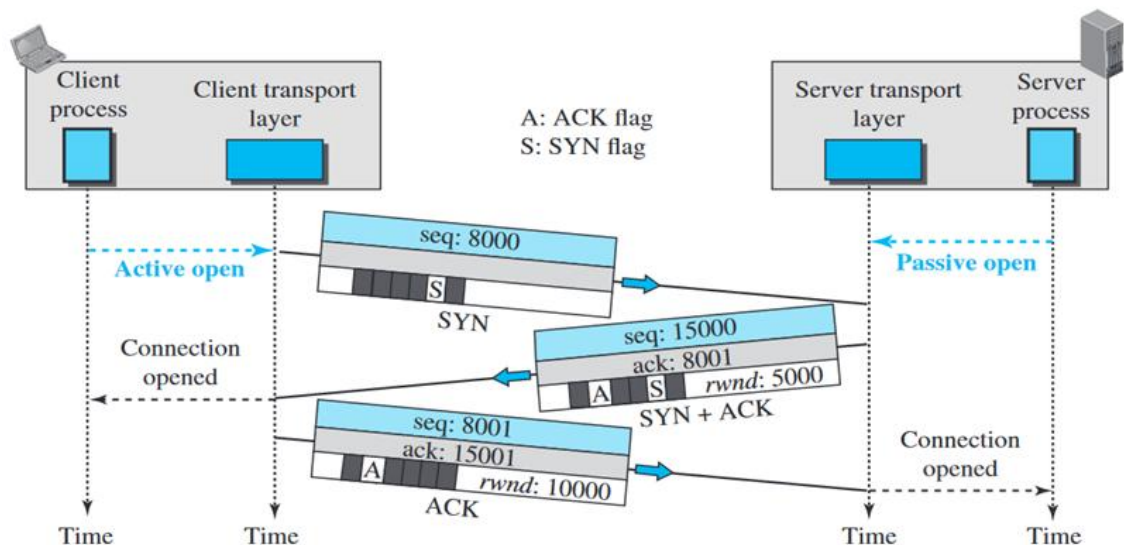


Fig. 4.7 TCP Connection Establishment

3. Explain working of Transport Layer Security (TLS) handshake protocol.

Client Hello:

The client sends a “Hello” message to the server containing supported TLS versions, cipher suites, and a random value.

Server Hello:

The server responds with its chosen TLS version, cipher suite, and its own random value.

Certificate Exchange:

The server sends its digital certificate (issued by a trusted CA) to prove its identity. It contains public key of the server.

Key or Pre Master Secret Exchange:

Client uses Servers public key to Encrypt pre master secret which is used to generate Session Key at both side. We can say that client and server exchanged session key logically.

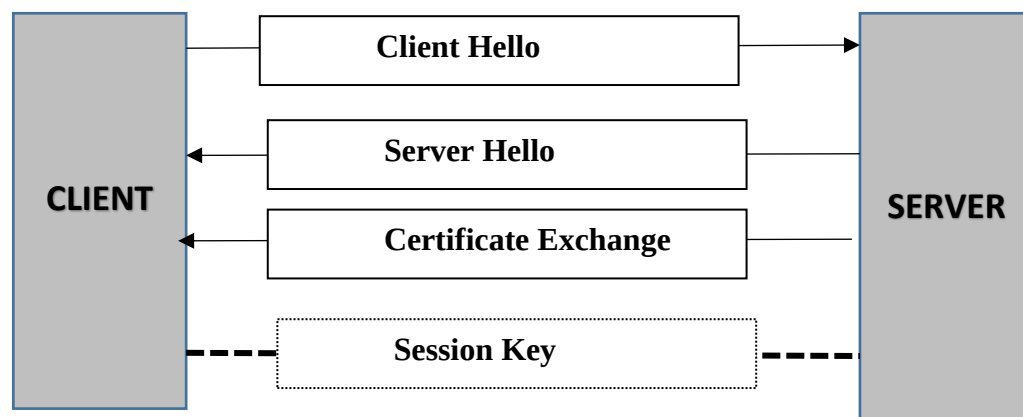


Fig. TLS Handshake Working

4. Explain Error Control in TCP.

Error Control in TCP

TCP provides reliability using error control.

- Error control includes mechanisms for detecting corrupted segments, lost segments, out-of-order segments, and duplicated segments.
- Error control also includes a mechanism for correcting errors after they are detected.

Error detection and correction in TCP is achieved through the use of three simple tools.

checksum

Each segment includes a checksum field which is used to check for a corrupted segment. If the segment is corrupted, it is discarded by the destination TCP and is considered as lost. TCP uses a 16-bit checksum that is mandatory in every segment.

Acknowledgment

TCP uses acknowledgments to confirm the receipt of data segments. Control segments that carry no data but consume a sequence number are also acknowledged. ACK segments are never acknowledged.

Retransmission

In modern implementations, a retransmission occurs

- If the retransmission timer expires **OR**
- Three duplicate ACK segments have arrived.

Retransmission based on Timers

The sender sends segments 1 and 2, which are acknowledged immediately by an ACK. Segment 3, however, is lost. The receiver receives segment 4, which is out of order. The receiver stores the data in the segment in its buffer but leaves a gap to indicate that there is no continuity in the data. The receiver immediately sends an acknowledgment to the sender, displaying the next byte it expects. Note that the receiver stores bytes 801 to 900, but never delivers these bytes to the application until the gap is filled.

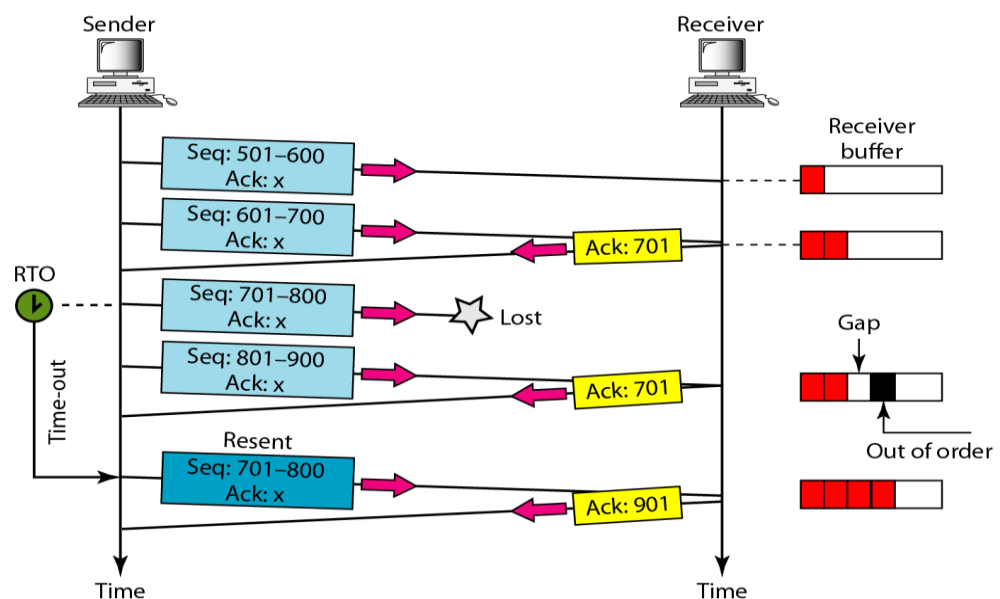


Fig. 4.12 Retransmission based on Timer

We have shown the timer for the earliest outstanding segment. The timer for this definitely runs out because the receiver never sends an acknowledgment for lost or out of-order segments. When the timer matures, the sending TCP resends segment 3, which arrives this time and is acknowledged properly. Note that the value in the second and third acknowledgments differs according to the corresponding rule.

Fast Retransmission based on 3-Duplicate ACK

In this scenario, we want to show the idea of fast retransmission. Our scenario is the same as the second except that the RTO has a higher value, In fast transmission even the timer not matured (expired) still after receiving three ACK segments for segment 3, sender will resend segment 3.

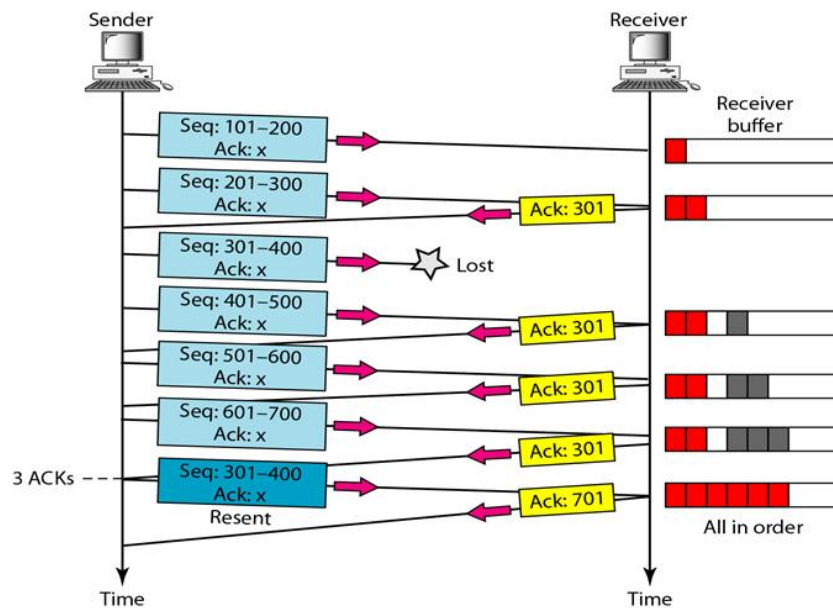


Fig. 4.11 Fast Retransmission

5. Differentiate between POP3 and IMAP.

Parameter	POP3	IMAP
Functionality	POP3 client downloads messages from server and then deleted them from server.	IMAP client Synchronize with server, allows users to access and manage their emails directly on the server
Storage	Less storage required at server.	Server need high storage capacity.
Offline Access	Yes	No
Email Management	No cannot manage mails on server.	Yes, emails can be stored in folders.
Uses	Suitable for Single device use.	Suitable for multiple device use.
Port No	110	143

6. Explain working of TELNET with help of diagram.

- TELNET is an abbreviation for TERminal NETwork. It is the standard TCP/IP protocol for virtual terminal service, which runs at port no 23.
- TELNET enables the establishment of a connection to a remote system in such a way that the local terminal appears to be a terminal at the remote system.
- There are two parties involved TELNET Client and TELNET server.

- TELNET client, which transforms the characters to a universal character set called network virtual terminal (NVT) characters and delivers them to the local TCP/IP protocol stack.

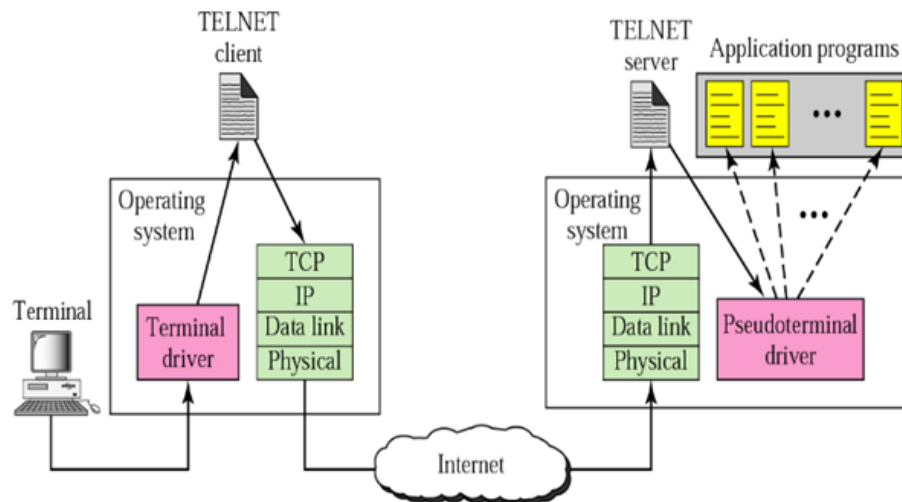


Fig. 5.20 TELNET Working

- The commands or text, in NVT form, travel through the Internet and arrive at the TCP/IP stack at the remote machine. Here the characters are delivered to the operating system and passed to the TELNET server, which changes the characters to the corresponding characters understandable by the remote computer, which is understandable by OS of remote machine.

7. Explain working of PGP with help of diagram.

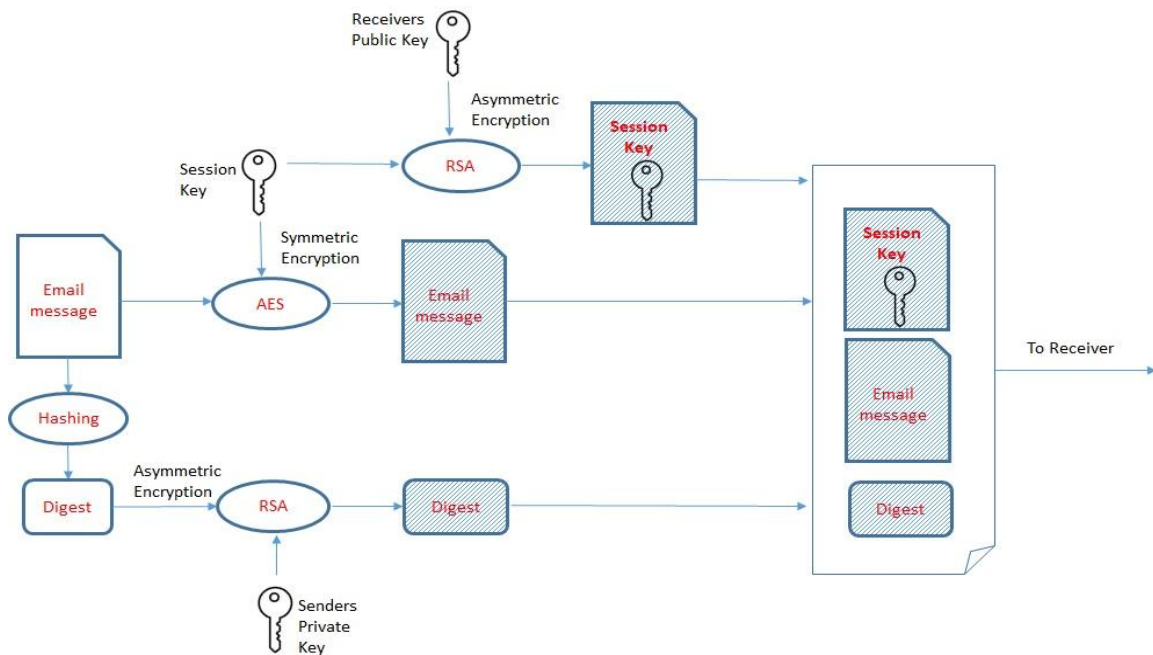
Pretty Good Privacy

Sender Site (Encryption Process):

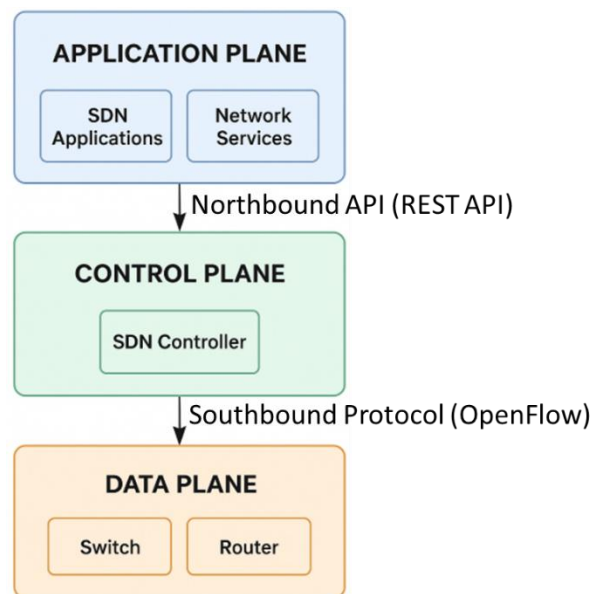
- The sender generates a random symmetric key (also called a session key) to encrypt the email using a fast symmetric encryption algorithm (e.g., AES).
- This session key is then encrypted with the recipient's public key using asymmetric encryption (e.g., RSA).
- Email is hashed and a digest is created, then email digest is encrypted using private key of sender.
- The encrypted session key, Email digest and the encrypted Email are sent to the recipient.

Receiver Site (Decryption Process)

- The recipient first uses their private key to decrypt the session key.
- The decrypted session key is then used to decrypt the actual email message. Recipient hash the email message to create email digest at receiver side.
- The recipient decrypt email digest received from sender using public key of sender, then compare email digest from sender with email digest generated at its own site, if match is found email is accepted otherwise email is discarded.



8. Explain SDN architecture.



SDN works by separating decision-making (control plane) from packet forwarding (data plane).

Applications tell the controller what they want, and the controller tells the switches, routers how to forward traffic, to complete requirements of Application plane

SDN architecture can be divided into three planes.

- Application Plane
- Control Plane
- Data Plane

Application plane

It doesn't directly control hardware — instead, it communicates with the SDN Controller (in the Control Plane) through northbound APIs.

The Application Plane talks to the Control Plane using Northbound APIs (usually REST APIs). These APIs are abstract, so the app developer doesn't need to know hardware details.

Example:

An app might just say:

“Ensure 10 Mbps guaranteed for Host A.”

The controller translates this into OpenFlow rules for the switches.

Control Plane

The control plane is responsible for making decisions about where and how packets should be forwarded.

In SDN, this logic is centralized in a controller (software), rather than being embedded in each network device.

Control plane installs flow rules into switches, routers and decides routing paths.

SDN Controller

A software platform (like ONOS, OpenDaylight, Ryu) that acts as the central decision-maker.

Example

Imagine a new device connects to the network. The switch doesn't know how to handle its packets, so it asks the SDN controller.

The controller decides, “Send these packets to Server A” and sends a flow rule to the switch.

Data Plane

The data plane (also called the forwarding plane) is responsible for moving packets based on rules given by the control plane.

Forwards, drops, or modifies packets according to flow table entries.

Example:

Once the controller installs a rule like

“Packets from Host X to Host Y → output port 3,”

the switch forwards those packets directly without asking the controller flag

9. Explain Voice-over Internet Protocol (VoIP), How VoIP works?

VoIP (Voice over Internet Protocol) is a technology that allows us to make voice calls using the Internet or an IP network instead of the traditional telephone network.

Examples include Internet-based calling in applications such as WhatsApp, Microsoft Teams, Zoom, and IP phones.

VoIP works in following steps

- i. Analog-to-Digital Conversion**
The sound is initially an analog signal. The VoIP device converts the analog voice signal into digital data.
- ii. Voice Compression**
The digital voice data is compressed using a codec.
A codec helps reduce the amount of data that needs to be transmitted.
- iii. Packetization**
The compressed voice is divided into small data packets.
For real-time voice transmission, RTP (Real-time Transport Protocol) packet is commonly used.
- iv. Transmission over the Internet**
The packets are transmitted through the IP network.
- v. Receiving the Packets**
The receiving device collects the incoming packets.
VoIP uses mechanisms such as sequence numbers and jitter buffers to help reconstruct the voice stream smoothly.
- vi. Decompression**
The codec at the receiving side converts the compressed voice data back into usable digital audio.
- vii. Digital-to-Analog Conversion**
The digital audio is converted back into an analog electrical/audio signal.
The speaker then produces the sound that Person B hears.

6marks

1. Explain in detail TCP Flow Control with the help of Receiver Window and Sender Window diagrams.

Flow Control in TCP

Flow control in TCP means sender must send data with an amount that can be handled by receiver without overwhelming.

Receive Window

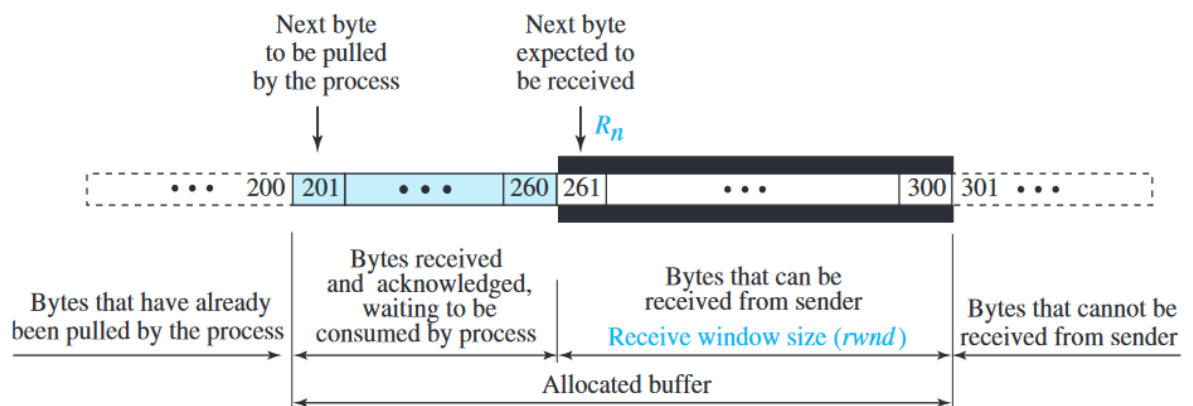


Fig. 4.12 Receive Window

The receiver advertises a window size i.e. **rwnd** (the amount of data receiver is ready to accept) to the sender. This size represents the free space in the receiver's buffer (*can be calculated by subtracting bytes not consumed the process from allocated buffer as shown in fig. 4.12*) and the sender must stay within this limit when transmitting data.

Send window

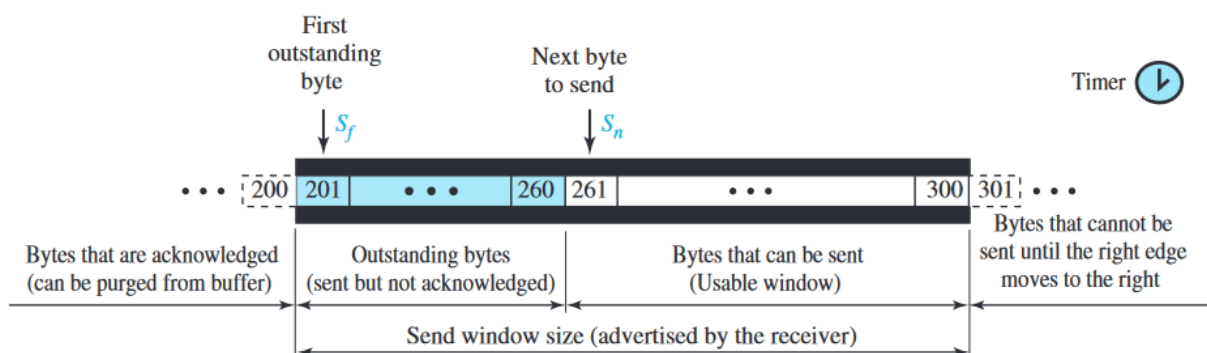


Fig. 4.13 Send Window

The sender adjusts its transmission rate based on the rwnd value received from the receiver.

If the *rwnd* is large, the sender can send more data at once. If it is small, the sender must slow down or stop until more space is available.

The sender keeps track of the amount of data that has been sent but not yet acknowledged (as shown in fig. 4.13) and ensures that this amount stays within the receiver's advertised window.

2. Define *cwnd*. Explain congestion control policies in TCP.

Congestion Window (*cwnd*)

- Use of *rwnd* guarantees that, there is no congestion at the other end, but there may be *congestion in the middle*. An intermediate router may receive data from more than one sender. No matter how large the buffers of a router may be, it may be overwhelmed with data, results congestion at middle.
- To control the number of segments to transmit, TCP uses another variable called a congestion window, *cwnd*, whose size is controlled by the congestion situation in the network (as we will explain shortly).
- The *cwnd* variable and the *rwnd* variable together define the size of the send window in TCP. The first is related to the congestion in the middle (network); the second is related to the congestion at the end. The actual size of the window is the minimum of these two.

Actual window size = minimum (*rwnd*, *cwnd*)

1. Slow Start – Exponential Increase

The Slow Start – Exponential Increase algorithm is based on the idea that the size of the congestion window (*cwnd*) starts with one maximum segment size (MSS), but it increases exponentially as follows.

Slow start: $\rightarrow cwnd = 2^0 = 1$

After Receiving **1** RTT $\rightarrow cwnd = 2^1 = 2$

After Receiving **2** RTT $\rightarrow cwnd = 2^2 = 4$

After Receiving **3** RTT $\rightarrow cwnd = 2^3 = 8$

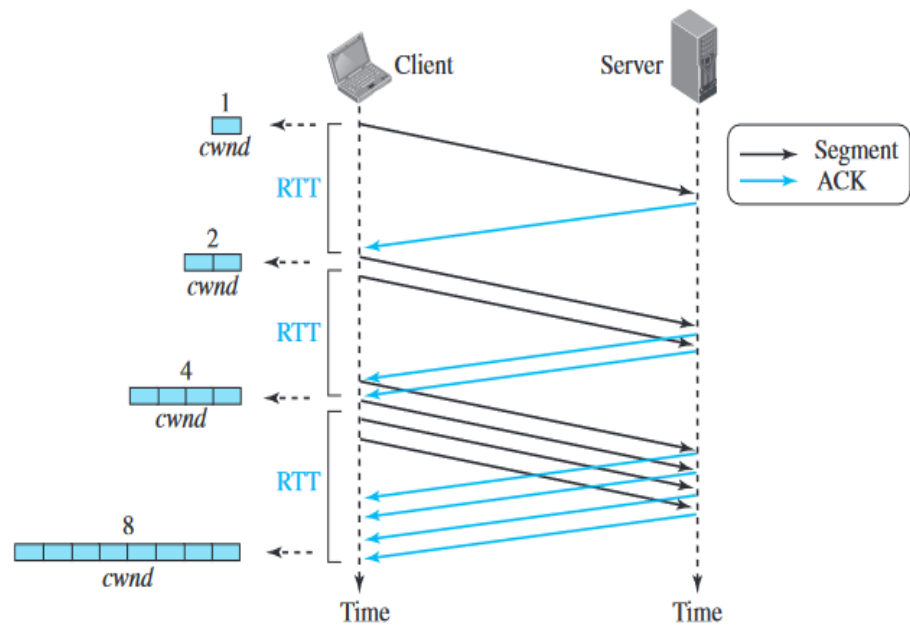


Fig. 4.14 Slow start – Exponential increase

2. Congestion Avoidance – additive increase

TCP defines another algorithm called congestion avoidance, which increases the *cwnd* additively instead of exponentially.

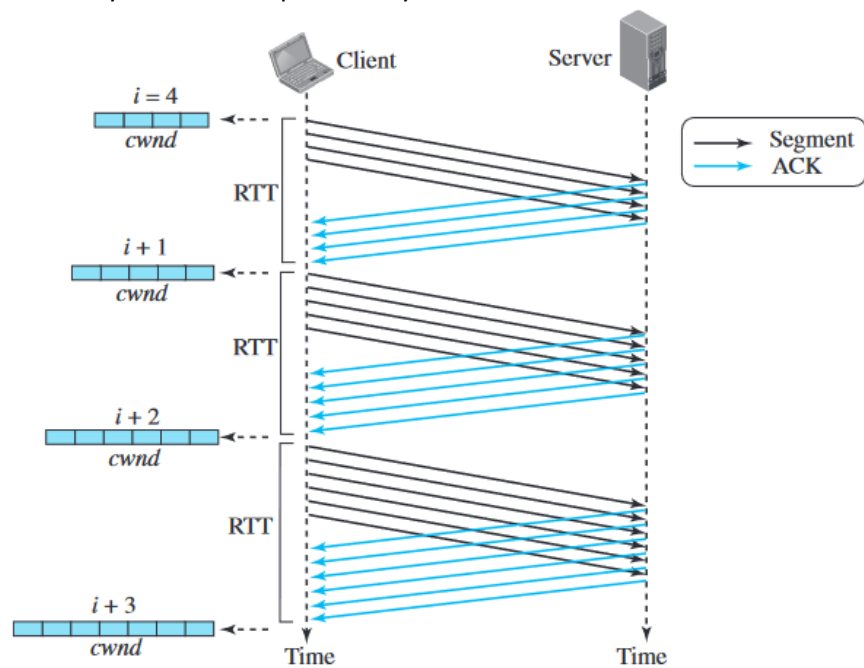


Fig. 4.15 Congestion avoidance – additive increase

3. Describe FTP. Explain communication over control connection and data connection in FTP.

File Transfer Protocol

- FTP uses the services of TCP. It needs two TCP connections.
- The well-known port **21** is used for the control connection and the well-known port **20** for the data connection.
- The control connection remains connected during the entire interactive FTP session.
- The data connection is opened and then closed for each file transferred.
- Control connection is there between client & server Control processes.
- Data connection is there between client & server Data transfer processes.

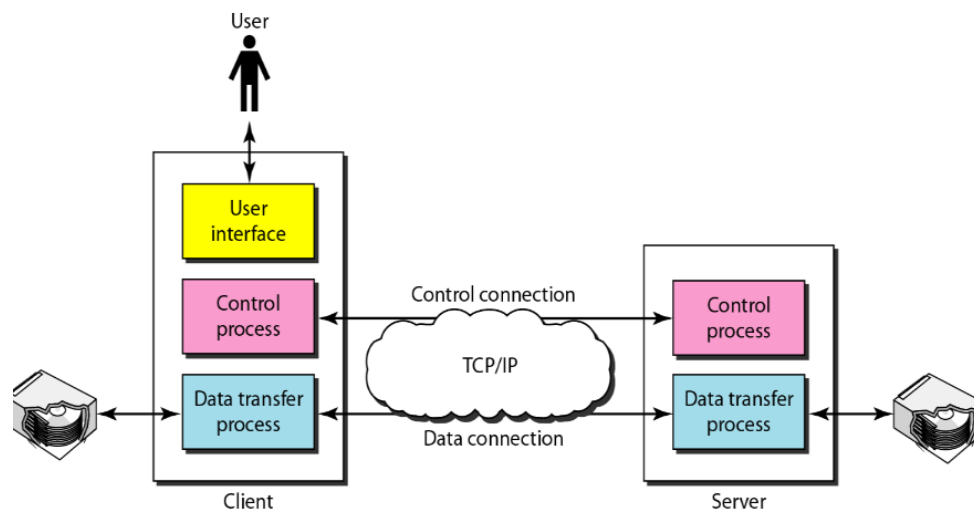


Fig. 5.8 File Transfer Protocol (FTP)

Communication over FTP Control Connection

- It uses the 7-bit ASCII character set (see Figure).
- Communication is achieved through commands and responses.
- Each command or response is only one short line, so we need not worry about file format or file structure.

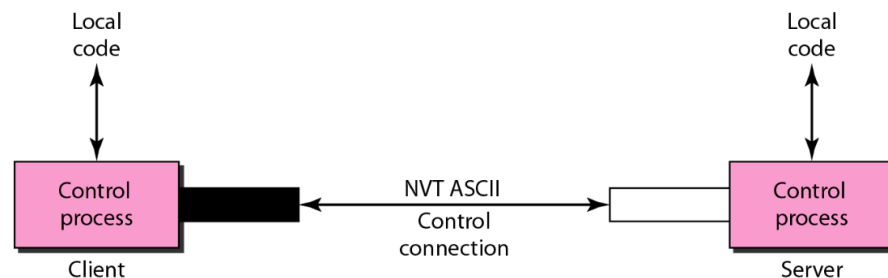
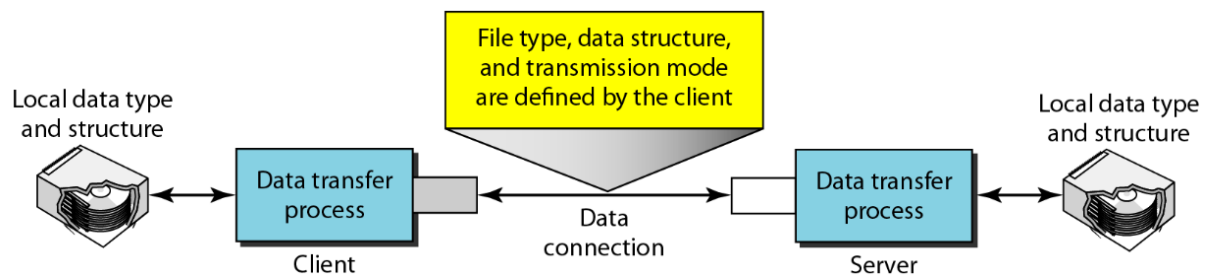


Fig. 5.9 FTP Control Connection

Communication over FTP Data Connection

- The purpose of the data connection is different from that of the control connection. It is used to transfer files.
- One of problem is that file structure at client and server may be differ, known as heterogeneity problem.
- The heterogeneity problem is resolved by defining three attributes of communication: **file type**, **data structure**, and **transmission mode**.



5.9 FTP Data Connection

4: List out SMTP commands. Use an example to show how a mail is sent using SMTP commands.

Command	Meaning / Purpose
HELO or EHLO	Greet the mail server and identify the client.
MAIL FROM:	Specifies the sender's email address.
RCPT TO:	Specifies the recipient's email address.
DATA	Indicates that the actual message (body + headers) will follow.
QUIT	Terminates the SMTP session.

SMTP uses a set of text-based commands between client and server to send emails in a reliable, step-by-step manner.

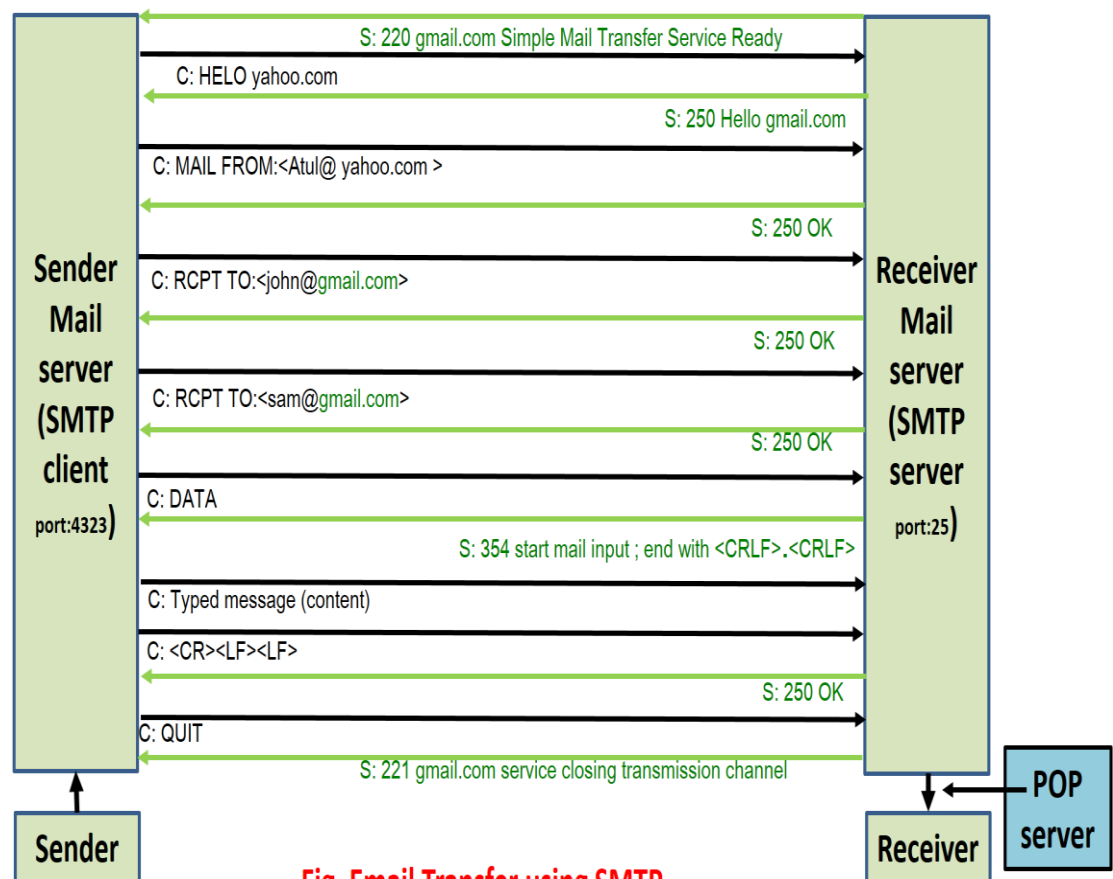


Fig. Email Transfer using SMTP

5. Illustrate the importance of Network Function Virtualization (NFV) with example.

Network Function Virtualization (NFV) is a technology in which traditional network functions such as routers, firewalls, load balancers, NAT, and VPN gateways are implemented as software running on standard servers instead of dedicated hardware appliances.

Importance of NFV

Importance	Explanation
1. Reduces cost	Reduces the need to purchase specialized networking hardware.
2. Flexible	Network functions can be created, modified, or removed through software.
3. Scalability	Additional virtual network functions can be started when traffic increases.
4. Faster deployment	A new firewall, router, or VPN can be deployed through software instead of installing new hardware.
5. Better resource utilization	Multiple network functions can share the same physical server resources.
6. Easy management	Network functions can be centrally monitored and managed.
7. Supports cloud computing	Network functions can run in data centers and cloud environments.
8. Hardware independence	Network services are less dependent on specialized vendor hardware.

6. Explain NFV architecture with its Applications.

Network Function Virtualization (NFV) is a modern approach to networking where traditional network services (like firewalls, load balancers, intrusion detection systems, etc.) that used to run on dedicated hardware devices are now implemented as software functions running on standard servers, storage, and switches.

Instead of buying special hardware for each function, NFV lets operators use virtualized network functions (VNFs) that run in a virtual environment.

NFV Architecture:

The NFV architecture is consists of three main components:

i. NFV Infrastructure (NFVI):

This is the foundation layer that provides the hardware and virtualization environment for VNFs.

It includes:

- Compute: Standard servers (CPU, RAM).
- Storage: Hard disks, SSDs.
- Network: Physical network to interconnect servers.

Virtualization Layer: Hypervisors (like VMware, KVM, Xen) or container runtimes (like Docker).
Basically, NFVI supplies the resources that VNFs need to run.

ii. Virtual Network Functions (VNFs):

- These are software-based implementations of traditional network functions.
- Each VNF performs a specific task such as firewalling, routing, NAT, or intrusion detection.
- Example: Virtual Firewall, Virtual Router, Virtual Load Balancer.

iii. NFV Management and Orchestration (MANO):

This is the "brain" of the architecture.

Responsible for:

- Deploying VNFs.
- Scaling VNFs up/down when traffic changes.
- Monitoring performance and failures.
- Managing resources in NFVI.

NFV Management and Orchestration			
Orchestrator (NFVO)	VNF Manager (VNFM)	Infrastructure Manager (VIM)	
NFV Infrastructure (NFVI)			
Compute	Storage	Network	Virtualization Layer
Virtualized Network Functions (VNFs)			
<u>vRouter</u>	<u>vFirewall</u>	<u>vLoad Balancer</u>	<u>vIDS</u>

Fig. NFV Architecture